

CODSOFT Python Programming Internship

Task 4: Rock Paper Scissors Game

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➤ Internship Domain:

Python Programming

Project

****Rock Paper Scissors Game Using Python****

Objective

The main goal of this project is to create a Rock Paper Scissors Game using Python. This game lets a user play against the computer. The computer makes a choice. The game figures out who wins based on the rules keeps track of scores and lets users play rounds.

Problem Statement

Rock Paper Scissors is a hand game that people play together. In this project the user plays against the computer.

The game should be able to:

- * Get the users input
- * Make a choice for the computer
- * Decide who wins
- * Keep track of scores
- * Let users play rounds
- * Show the final result

Tools and Technologies Used

Tool	Purpose
Python 3	Programming Language
VS Code / IDLE	Code Editor
Command Prompt / Terminal	Program Execution
Random Module	Random Computer Choice

System Requirements

- **Hardware Requirements**
 - * Computer/Laptop
 - * Minimum 2 GB RAM
- **Software Requirements**
 - * Python 3.x
 - * Any Code Editor (VS Code, PyCharm, IDLE)

Algorithm

- **Step 1**

Make a list with:

- * Rock

- * Paper

- * Scissors

- **Step 2**

Set the users score and the computers score to 0.

- **Step 3**

Get the users choice.

- **Step 4**

Make a choice for the computer.

- **Step 5**

Compare the two choices. Decide:

- * Who wins

- * Who loses

- * If it's a tie

- **Step 6**

Update the score.

- **Step 7**

Show the score.

- **Step 8**

Ask the user if they want to play

- **Step 9**

Show the score and who wins when the game is over.

Source Code

****Rock Paper Scissors Game Using Python****

```
import random

print("==== Rock Paper Scissors Game ====")

user_score = 0
computer_score = 0

choices = ["rock", "paper", "scissors"]

while True:

    print("\nChoose one:")
    print("Rock")
    print("Paper")
    print("Scissors")

    user_choice = input("Enter your choice: ").lower()

    if user_choice not in choices:
        print("Invalid choice! Please select Rock, Paper, or Scissors.")
        continue

    computer_choice = random.choice(choices)
```

```
print("\nYour Choice:", user_choice.capitalize())

print("Computer Choice:", computer_choice.capitalize())


# Winner Logic

if user_choice == computer_choice:

    print("Result: It's a Tie!")


elif (

    (user_choice == "rock" and computer_choice == "scissors") or

    (user_choice == "paper" and computer_choice == "rock") or

    (user_choice == "scissors" and computer_choice == "paper")

):

    print("Result: You Win!")

    user_score += 1


else:

    print("Result: Computer Wins!")

    computer_score += 1


# Score Board

print("\n===== Score Board =====")

print("Your Score:", user_score)

print("Computer Score:", computer_score)


# Play Again

play_again = input("\nDo you want to play again? (yes/no): ").lower()


if play_again != "yes":
```

```
print("\n===== Final Score =====")
print("Your Score:", user_score)
print("Computer Score:", computer_score)

if user_score > computer_score:
    print("Overall Winner: You")
elif computer_score > user_score:
    print("Overall Winner: Computer")
else:
    print("Overall Result: Tie")

print("\nThanks for playing!")
break
```

- **Explanation of the Code**

Random Module

The random module is used to make the computers choice.

User Input

The user enters either:

- * Rock

- * Paper

- * Scissors

Winner Logic

The program checks:

- * Rock beats Scissors

* Scissors beats Paper

* Paper beats Rock

Score Tracking

The game keeps track of:

* User Score

* Computer Score

Play Again Feature

Users can play rounds without restarting the game.

Final Result

When the user exits the game the program shows the scores and announces the overall winner.

- **Sample Output**

===== Rock Paper Scissors Game =====

Choose one:

Rock

Paper

Scissors

Enter your choice: rock

Your Choice: Rock

Computer Choice: Scissors

Result: You Win!

===== **Score Board** =====

Your Score: 1

Computer Score: 0

Do you want to play ? (y/n): y

Final Score Example

===== Final Score =====

Your Score: 3

Computer Score: 2

Overall Winner: You

Thanks for playing!

Features

- * User vs Computer Gameplay
- * Random Computer Choice
- * Winner Determination Logic
- * Score Tracking System
- * Multiple Rounds
- * Play Again Feature
- * Input Validation
- * Final Winner Declaration

Advantages

- * Easy to understand and play
- * Interactive user experience
- * Demonstrates game development concepts
- * Improves logical thinking and decision-making skills
- * Beginner-friendly Python project

Learning Outcomes

Through this project I learned about:

- * Python Basics
- * Lists
- * Loops
- * Conditional Statements
- * User Input Handling
- * Random Number Generation
- * Score Tracking
- * Game Logic Implementation

Future Enhancements

- * Graphical User Interface (GUI)
- * Multiplayer Mode
- * Best of 3 / Best of 5 Mode
- * Sound Effects
- * Game Statistics
- * Leaderboard System

#Conclusion

The Rock Paper Scissors Game was successfully developed using Python. The project demonstrates the use of loops, conditional statements, lists, user input handling, and random number generation. The score tracking and multiple-round functionality make the game more interactive and user-friendly.

Result

The Rock Paper Scissors Game was successfully implemented and tested. The application allows users to play multiple rounds against the computer, track scores, and view the final winner. The game provides an interactive way to understand Python programming concepts such as loops, conditional statements, lists, and random number generation.